

# Three-tier process design decisions

Airbnb NYC Track 1 · Group D · Machine Learning module · University of Essex Online · April 2026 intake  
Author: Ariel Mella · v1.3 — Minimum EDA, Regression and Clustering Register

## 1. Tier system

| Tier      | Meaning                                          | Use here                                                                                                                                                                 |
|-----------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T1</b> | Academic / taught-unit support                   | Core ML logic: regression, polynomial features, K-Means, validation metrics, silhouette and claim limits.                                                                |
| <b>T2</b> | Tool documentation / accepted Python ML practice | scikit-learn / script choices: split, StandardScaler, LinearRegression, KMeans and metrics.                                                                              |
| <b>T3</b> | Project judgement / dataset constraint           | Used only when T1/T2 cannot decide the value: cleaned dataset constraint, weak demand proxies, K = 3 interpretation, listing-not-customer segmentation and claim weight. |

## 2. Cleaning, EDA and feature basis

| Area / question     | Choice / value                                                                                                                                                        | Tier           | Minimum rationale and claim limit                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cleaned input       | 48,895 rows → 48,847 listings                                                                                                                                         | <b>T1 / T3</b> | Invalid / unusable records are removed, mainly for price. The cleaned set stays close to the source because this is a descriptive Track 1 model, not a production rebuild (Dgomonov, 2019; James et al., 2023). |
| Missingness / zeros | Modelling fields complete; zeros kept                                                                                                                                 | <b>T1 / T3</b> | availability_365 = 0 and zero reviews can be real signals. They are kept, but read carefully: zero availability is not proof of demand, and zero reviews is not proof of no use.                                |
| Outliers            | log_price; minimum_nights q99 cap = 45                                                                                                                                | <b>T1 / T3</b> | Price is right-skewed and minimum_nights has extreme values. Log price and q99 cap reduce leverage without deleting the market shape (Tukey, 1977; James et al., 2023).                                         |
| EDA signals         | Room type, neighbourhood group, price distribution, availability and reviews                                                                                          | <b>T1 / T3</b> | EDA shows a tiered and long-tailed market. Room type and neighbourhood matter; availability and reviews suggest activity. This stays descriptive, not causal.                                                   |
| Feature basis       | Regression: type, borough, coordinates, nights, reviews, availability, host count. Clustering: log_price, nights, reviews_per_month, availability, room_type one-hot. | <b>T1 / T3</b> | The features cover listing type, location, activity, availability and host scale. Claim limit: no amenities, photos, bookings, revenue, calendar demand or guest/customer data.                                 |
| Dataset limits      | Historic 2019 listing snapshot                                                                                                                                        | <b>T3</b>      | Good for pricing patterns and listing segmentation. Not enough for current NYC claims, causal price impact or true customer segmentation.                                                                       |

### 3. Regression rationale

| Area / question   | Choice / value                                                                                                       | Tier           | Minimum rationale and claim limit                                                                                                                                                                                                                                       |
|-------------------|----------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regression target | log_price                                                                                                            | <b>T1 / T2</b> | Prices are right-skewed. The log target makes the model more stable than raw price and easier to read (Bishop, 2006; James et al., 2023). Claim limit: transformed price patterns, not causal price impact.                                                             |
| Test basis        | 80/20 split; random_state = 1; same held-out test set                                                                | <b>T2</b>      | All models are compared on the same split, so $R^2$ and RMSE move on a like-for-like basis. Fixed seed makes the run repeatable (scikit-learn train_test_split; James et al., 2023). Limitation: k-fold is better future validation.                                    |
| Model sequence    | Room-type baseline → multiple linear → degree-2 polynomial                                                           | <b>T1 / T3</b> | Progresses from simple to more expressive while staying explainable. Evidence: $R^2$ 0.372 → 0.480 → 0.534; RMSE 0.548 → 0.498 → 0.471. Linear/polynomial regression are standard (James et al., 2023; Hastie et al., 2009).                                            |
| Features used     | Nine available numeric fields: type, borough code, coordinates, nights, reviews, availability and host-listing count | <b>T1 / T3</b> | Features map to price, location, listing type, activity and host scale. reviews_per_month and availability_365 are weak activity / demand signals, not booking demand. Constraint: AB_NYC_2019 has no amenities, photos, bookings or live demand data (Dgomonov, 2019). |
| Scaling           | StandardScaler for multiple and polynomial models; fit on training data only                                         | <b>T1 / T2</b> | Scaling supports polynomial terms and avoids test leakage when fitted only on training data (scikit-learn StandardScaler; James et al., 2023).                                                                                                                          |
| VIF check         | Max VIF = 1.625; no automatic feature removal                                                                        | <b>T1</b>      | VIF checks coefficient instability from overlapping features. Current run is below the common 5.0 threshold, so all features stay. I keep business-meaningful variables rather than auto-pruning without need (James et al., 2023).                                     |
| Claim weight      | Guidance, not Smart Pricing automation                                                                               | <b>T1 / T3</b> | $R^2$ around 0.53 is useful for broad guidance but not enough for automated pricing. The dataset is observational, so the report avoids causal pricing claims (Crawford, 2006; Chernozhukov, 2024).                                                                     |

## 4. Clustering rationale

| Area / question   | Choice / value                                                                                 | Tier           | Minimum rationale and claim limit                                                                                                                                                                                               |
|-------------------|------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Segmentation unit | Listing segmentation, not customer segmentation                                                | <b>T3</b>      | AB_NYC_2019 contains listings and hosts, but no guest or booking-level customer records. So the valid unit is the listing. Claim limit: do not call this true customer segmentation.                                            |
| Method            | K-Means                                                                                        | <b>T1 / T2</b> | K-Means is the taught method for this unit and is easy to profile for a business audience (Bishop, 2006; scikit-learn KMeans).                                                                                                  |
| Features used     | log_price, minimum_nights, reviews_per_month, availability_365 and room_type one-hot variables | <b>T1 / T3</b> | Features cover price, stay rules, activity, availability and listing type. One-hot avoids a fake order between room types. Source run uses drop_first=True; full-dummy sensitivity is cleaner future work.                      |
| Scaling           | StandardScaler before K-Means                                                                  | <b>T1 / T2</b> | K-Means is distance-based, so unscaled large-range variables would dominate the clusters (James et al., 2023; scikit-learn StandardScaler).                                                                                     |
| K diagnostics     | K = 2 to 6; inertia / SSE and silhouette                                                       | <b>T1 / T2</b> | Elbow / SSE and silhouette are accepted cluster-validity checks. In this run inertia bends around K = 2-3, while silhouette rises to K = 6 (Rousseeuw, 1987; Tippaya Thinsungnoen et al., 2015; scikit-learn silhouette_score). |
| K used            | K = 3                                                                                          | <b>T1 / T3</b> | K = 3 balances diagnostics and interpretation. Higher K may improve silhouette but creates more micro-segments than the report can explain cleanly. Current profiles: Private-room 45.1%, Entire-home 52.6%, Shared-room 2.4%.  |
| Segment labels    | Private-room, Entire-home and Shared-room segments                                             | <b>T3</b>      | Labels come from room_type cross-tabs: the clusters are dominated by one room type each. Claim limit: labels are working business labels, not ground truth classes.                                                             |
| Visual checks     | Silhouette-by-segment plot and price/availability scatter                                      | <b>T2 / T3</b> | The plots test whether K = 3 is usable and readable. They support the story but do not prove permanent market categories. Alternative clustering remains future work.                                                           |

## 5. Claim-weight guardrails

| Guardrail                    | Chosen stance                                                                                                                                     | Tier           | Minimum rationale                                                                                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fairness                     | No fairness claim                                                                                                                                 | <b>T1 / T3</b> | The data is listing-level and has no guest/host demographic data for a real fairness audit. Operational use would need separate fairness data and controls (Leslie, 2019; Corrêa et al., 2023). |
| Transparency                 | Show R <sup>2</sup> /RMSE, VIF, K diagnostics, profiles and logs                                                                                  | <b>T1 / T2</b> | The report is short and executive-facing. Evidence needs to be visible and explainable, not hidden behind a black-box style presentation (Leslie, 2019; scikit-learn documentation).            |
| Responsible use              | Decision support only; human review stays in the loop                                                                                             | <b>T1 / T3</b> | Historic data, missing business drivers and modest predictive strength mean the model should guide discussion, not set prices or automate host actions (Leslie, 2019).                          |
| What can / cannot be claimed | Can: pricing patterns and listing segments. Cannot: causal impact, current NYC market, true customer segmentation or automated pricing readiness. | <b>T1 / T3</b> | Claims stay descriptive and scoped. Stronger claims need new data, causal design and more validation (Crawford, 2006; Chernozhukov, 2024).                                                      |

## 6. Core sources

Bishop, C.M. (2006) *Pattern Recognition and Machine Learning*. Springer.

Chernozhukov, V. (2024) *Applied Causal Inference Powered by ML and AI*. arXiv.

Corrêa, N.K. et al. (2023) Worldwide AI ethics: A review of 200 guidelines and recommendations for AI governance. *Patterns*, 4(10).

- Crawford, S.L. (2006) Correlation and Regression. *Circulation*, 114(9).
- Dgomonov (2019) *New York City Airbnb Open Data*. Kaggle.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009) *The Elements of Statistical Learning*. Springer.
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer.
- Leslie, D. (2019) *Understanding artificial intelligence ethics and safety*. The Alan Turing Institute.
- Rousseeuw, P.J. (1987) Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, pp. 53–65.
- Tippaya Thinsungnoen et al. (2015) The Clustering Validity with Silhouette and Sum of Squared Errors. *Proceedings of the 3rd International Conference on Industrial Application Engineering*.
- Tukey, J.W. (1977) *Exploratory Data Analysis*. Addison-Wesley.
- scikit-learn documentation: `train_test_split`, `LinearRegression`, `PolynomialFeatures`, `StandardScaler`, `KMeans`, `silhouette_score`, `R2 score` and `mean_squared_error`.